SI Session Planning Form

SI Leader: Brandon Cobb
Session Week: 3
Session#: 3
Instructor: Professor Tramontozzi
Old Plans: SharePoint
Session Goal: Students will have reviewed unit conversions, algebraic manipulation of exact and inexact numbers, calculating their significant figures after multi-step operations like addition and multiplication. Student will review how to perform measurements using measurment devices with fixed uncertainty values, record data with units and write scienticific notation. Students are given a reference sheet for conversion factors.

Opener

Take Attendance

Collaborative Learning Techniques	Clusters
Learning Strategy	Send a Problem
Time	12:00—12:10
Materials	A writing utensil and index cards.
Lecture/Textbook/Old Plans/Slide References	Class roster, Chapter 1: Basics of Matter, Chapter 2: Measurments

Definitions

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1. **Clusters:** In Clusters, group participants are divided into smaller groups for discussion. They may also be allowed to self-select the small group they want to be in. After discussing the assigned topic, the cluster may report their findings to the large group.

2. **Send a Problem:** Generate a list of problems. Assign a different problem to each student. Give the students a minute to complete the first step of their assigned problem. After the minute, students pass their problem to the students to their right, who then complete the second step of the problem. Continue passing the problem until the steps are completed. After, have the students present their originally-assigned problems with the completed solutions to the entire group.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

Send a Problem (Chemistry Ice Breaker): Generate a list of chemistry problems of varying difficulty. Assign each student a different problem to solve. Give students one minute to complete the first step of their assigned problem. After the minute, each student passes their problem to the group on their right. The receiving group then completes the next step of the problem. This continues until all steps are completed. Finally, each student presents their originally-assigned problem along with the completed solution to the entire class.

1. The SI leader pairs up students based on proximity seating. If a student does not have a partner, the leader joins them.
2. Each pair is assigned a small chemistry task (e.g., determining concentration, calculating molar mass, predicting reaction products) suitable for a first step in a multi-step problem.
3. Pairs have two minutes to decide on their approach and collect any necessary tools or reference materials provided by the leader.
4. Pairs complete the first step of their problem, recording results clearly, and then pass the problem and any physical materials (if applicable) to the next group.
5. The next group evaluates the previous step and completes the subsequent step. If a group identifies an incorrect approach or calculation, they are allowed to correct it for the next step.
6. After all steps are completed, each original problem owner presents the full, completed solution to the class, highlighting any corrections or insights from peer contributions.
7. The SI leader facilitates discussion on problem-solving strategies, peer collaboration, and common chemistry misconceptions.

Students engage collaboratively, seeing multiple approaches to the same problem, learning from their peers, and practicing stepwise problem-solving on a shared centralized canvas of the problem.

Content (fully worked out problems answers)

(1) A chemist pours 0.375 L into a beaker already containing 125 mL. Total volume in liters?

a. 0.500 L

b. 0.250 L

c. 0.375 L

d. 0.125 L

Solution:

$$125 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.125 \text{ L}$$

$$V_{\text{total}} = 0.375 \text{ L} + 0.125 \text{ L} = 0.500 \text{ L}$$

Answer: a. 0.500 L

Content (fully worked out problems & answers) continued...

(2) Block of mass 540 g, volume half of 10. cm cube. Density?

a. 1.1 g/cm^3

b. 0.54 g/cm^3

c. 2.0 g/cm^3

d. 1.08 g/cm^3

Solution:

$$V_{\text{cube}} = 10^3 = 1000 \text{ cm}^3$$

$$V_{\text{metal}} = \frac{1}{2} \times 1000 = 500 \text{ cm}^3$$

$$\rho = \frac{m}{V} = \frac{540 \text{ g}}{500 \text{ cm}^3} = 1.1 \text{ g/cm}^3$$

Answer: a. 1.1 g/cm^3

Content (fully worked out problems & answers) continued...

(3) 1.2 kg water poured into 2 equal beakers. Mass per beaker?

a. 6.0×10^2 g

b. 6000 g

c. 1.2 g

d. 6.0×10^1 g

Solution:

$$1.2 \text{ kg} \times 1000 \ \frac{\text{g}}{\text{kg}} = 1200 \text{ g}$$

$$m_{\text{per beaker}} = \frac{1200 \text{ g}}{2} = 6.0 \times 10^2 \text{ g}$$

Answer: a. 6.0×10^2 g

Content (fully worked out problems & answers) continued...

(4) Oil of 170 g, density 0.85 g/mL. Volume in liters?

a. 0.20 L

b. 0.170 L

c. 2.00 L

d. 0.020 L

Solution:

$$V = \frac{m}{\rho} = \frac{170 \text{ g}}{0.85 \text{ g/mL}} = 200 \text{ mL}$$

$$V_{\text{L}} = 200 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.20 \text{ L}$$

Answer: a. 0.20 L

Content (fully worked out problems & answers) continued...

(5) Cylinder 250 mL alcohol; 0.30 kg water added. Final volume in liters?

a. 0.55 L

b. 0.35 L

c. 0.75 L

d. 0.25 L

Solution:

$$0.30 \text{ kg} \times 1000 \frac{\text{g}}{\text{kg}} = 300 \text{ g}$$

$$V_{\text{water}} = \frac{300 \text{ g}}{1 \text{ g/mL}} = 300 \text{ mL}$$

$$V_{\text{total}} = 250 \text{ mL} + 300 \text{ mL} = 550 \text{ mL}$$

$$V_{\text{L}} = 550 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.55 \text{ L}$$

Answer: a. 0.55 L

Content (fully worked out problems & answers) continued...

(6) Beaker contains 0.5 kg, half the required mass. Required mass in grams?

a. 1000 g

b. 500 g

c. 250 g

d. 2000 g

Solution:

$$m_{\text{required}} = 0.5 \text{ kg} \times 2 = 1.0 \text{ kg}$$

$$1.0 \text{ kg} \times 1000 \ \frac{\text{g}}{\text{kg}} = 1000 \text{ g}$$

Answer: a. 1000 g

Content (fully worked out problems answers) continued...

(7) A student measures the mass of a sample as 12.5 g, but the true mass is 0.125 kg. What is the correct mass in grams?

a. 125 g

b. 12.5 g

c. 0.125 g

d. 1.25 g

Solution:

$$0.125 \text{ kg} \times 1000 \frac{\text{g}}{\text{kg}} = 125 \text{ g}$$

Answer: a. 125 g

Content (fully worked out problems & answers) continued...

(8) Rectangular prism: volume twice mass in cm^3, mass 120 g. Density?

a. 0.50 g/cm^3

b. 1.00 g/cm^3

c. 0.25 g/cm^3

d. 2.00 g/cm^3

Solution:

$$V = 2 \times 120 \text{ cm}^3 = 240 \text{ cm}^3$$

$$\rho = \frac{m}{V} = \frac{120 \text{ g}}{240 \text{ cm}^3} = 0.50 \text{ g/cm}^3$$

Answer: a. 0.50 g/cm^3

Content (fully worked out problems & answers) continued...

(9) Mix 300. mL (0.90 g/mL) with 200. mL (1.2 g/mL). Average density?

a. 1.0 g/mL

b. 0.98 g/mL

c. 1.10 g/mL

d. 0.90 g/mL

Solution:

$$m_A = 300 \text{ mL} \times 0.9 \text{ g/mL} = 270 \text{ g}$$

$$m_B = 200 \text{ mL} \times 1.2 \text{ g/mL} = 240 \text{ g}$$

$$V_{\text{total}} = 300. \text{ mL} + 200. \text{ mL} = 500. \text{ mL}$$

$$\rho_{\text{avg}} = \frac{m_{\text{total}}}{V_{\text{total}}} = \frac{270 + 240}{500.} = 1.0 \text{ g/mL}$$

Answer: a. 1.0 g/mL

Content (fully worked out problems & answers) continued...

(10) Flask contains 0.6 L; student removes 200 mL. Remaining volume in liters?

a. 0.4 L

b. 0.2 L

c. 0.8 L

d. 0.6 L

Solution:

$$200 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.2 \text{ L}$$

$$V_{\text{remaining}} = 0.6 \text{ L} - 0.200 \text{ L} = 0.4 \text{ L}$$

Answer: a. 0.4 L

Content (fully worked out problems & answers) continued...

(11) Metal mass 300 g, scale reads 0.3 kg. Correct grams?

a. Yes, 300 g

b. No, 30 g

c. No, 3 g

d. No, 3000 g

Solution:

$$0.3 \text{ kg} \times 1000 \frac{\text{g}}{\text{kg}} = 300 \text{ g}$$

Answer: a. Yes, 300 g

Content (fully worked out problems & answers) continued...

(12) Rectangular prism: 10 cm 5 cm, mass 250 g, density 2.5 g/cm^3. Height?

a. 2 cm

b. 5 cm

c. 1 cm

d. 10 cm

Solution:

$$V = \frac{m}{\rho} = \frac{250 \text{ g}}{2.5 \text{ g/cm}^3} = 100 \text{ cm}^3$$

$$h = \frac{V}{l \times w} = \frac{100 \text{ cm}^3}{10 \text{ cm} \times 5 \text{ cm}} = 2 \text{ cm}$$

Answer: a. 2 cm

Content (fully worked out problems & answers) continued...

(13) 50 mL liquid (0.8 g/mL); 25 mL evaporates. Mass left?

a. 20 g

b. 40 g

c. 10 g

d. 25 g

Solution:

$$V_{\text{left}} = 50 \text{ mL} - 25 \text{ mL} = 25 \text{ mL}$$

$$m = \rho \times V_{\text{left}} = 0.8 \text{ g/mL} \times 25 \text{ mL} = 20 \text{ g}$$

Answer: a. 20 g

Content (fully worked out problems & answers) continued...

(14) 2.000 moles HCl, molar mass 36.46 g/mol. Mass should be?

a. 72.92 g

b. 36.46 g

c. 18.23 g

d. 144.12 g

Solution:

$$m = n \times M = 2.000 \text{ mol} \times 36.46 \text{ g/mol} = 72.92 \text{ g}$$

Answer: a. 72.92 g

Content (fully worked out problems & answers) continued...

(15) 0.5 L NaOH, add 250 mL, container reads 0.8 L. Consistent?

a. Yes

b. No, it should read 0.7 L

c. No, it should read 0.75 L

d. No, it should read 1.0 L

Solution:

$$0.250 \text{ L} + 0.5 \text{ L} = 0.75 \text{ L}$$

Answer: a. Yes

Content (fully worked out problems & answers) continued...

(16) Cube of ice edge 4.00 cm, density 0.920 g/cm^3. Mass?

a. 58.9 g

b. 64.0 g

c. 56.0 g

d. 60.0 g

Solution:

$$V = 4.00^3 \text{ cm}^3 = 64.0 \text{ cm}^3$$

$$m = \rho \times V = 0.920 \text{ g/cm}^3 \times 64.0 \text{ cm}^3 = 58.88 \text{ g} \approx 58.9 \text{ g}$$

Answer: a. 58.9 g

Content (fully worked out problems & answers) continued...

(17) Sample contains 0.03 mol Mg (24.31 g/mol). Mass?

a. 0.73 g

b. 7.29 g

c. 0.073 g

d. 0.30 g

Solution:

$$m = n \times M = 0.03 \text{ mol} \times 24.31 \text{ g/mol} = 0.7293 \text{ g} \approx 0.73 \text{ g}$$

Answer: a. 0.73 g

Content (fully worked out problems & answers) continued...

(18) A student heats 30 g of ice. How many grams of water are formed?

a. 30 g

b. 0 g

c. 15 g

d. 60 g

Solution:
All the ice melts, so the mass of water is equal to the mass of ice:

$$\text{Mass of water} = 30 \text{ g}$$

Answer: a. 30 g

Content (fully worked out problems & answers) continued...

(19) 0.900 kg ethanol, density 0.789 g/mL. Volume in mL?

a. 1140 mL

b. 900 mL

c. 700 mL

d. 1200 mL

Solution:

$$0.900 \text{ kg} \times 1000 \ \frac{\text{g}}{\text{kg}} = 900. \text{ g}$$

$$V = \frac{m}{\rho} = \frac{900. \text{ g}}{0.789 \text{ g/mL}} \approx 1140 \text{ mL}$$

Answer: a. 1140 mL

Content (fully worked out problems & answers) continued...

(20) Mix 150. mL (1.10 g/mL) with 50. mL (0.90 g/mL). Average density?

a. 1.05 g/mL

b. 1.00 g/mL

c. 0.95 g/mL

d. 1.10 g/mL

Solution:

$$m_1 = 150.\ \text{mL} \times 1.10\ \text{g/mL} = 165\ \text{g},$$

$$m_2 = 50.\ \text{mL} \times 0.90\ \text{g/mL} = 45\ \text{g}$$

$$V_{\text{total}} = 150\ \text{mL} + 50\ \text{mL} = 200\ \text{mL}$$

$$\rho_{\text{avg}} = \frac{m_{\text{total}}}{V_{\text{total}}} = \frac{165 + 45}{200} = 1.05\ \text{g/mL}$$

Answer: a. 1.05 g/mL

Content (fully worked out problems & answers) continued...

(21) 300. mL water, 5% evaporates. Remaining volume in mL and L?

a. 285 mL = 0.285 L

b. 290 mL = 0.290 L

c. 250 mL = 0.250 L

d. 300 mL = 0.300 L

Solution:

$$V_{\text{evap}} = 0.05 \times 300.\ \text{mL} = 15\ \text{mL}$$

$$V_{\text{left}} = 300.\ \text{mL} - 15\ \text{mL} = 285\ \text{mL}$$

$$V_{\text{left}} = 285\ \text{mL} \times \frac{1\ \text{L}}{1000\ \text{mL}} = 0.285\ \text{L}$$

Answer: a. 285 mL = 0.285 L

Content (fully worked out problems & answers) continued...

(22) Block mass 200 g, density 0.8 g/cm^3. Student thinks volume 300 cm^3. Correct volume?

a. 300 cm^3

b. 250 cm^3

c. 200 cm^3

d. 400 cm^3

Solution:

$$V = \frac{m}{\rho} = \frac{200 \text{ g}}{0.8 \text{ g/cm}^3} = 300 \text{ cm}^3$$

Answer: a. 300 cm^3

Content (fully worked out problems answers) continued...

(23) A chemist adds 58.44 g of NaCl to a beaker containing 500. g of water. What is the total mass of the solution?

a. 558.44 g

b. 542.00 g

c. 500.00 g

d. 58.44 g

Solution:

$$\text{Total mass} = \text{mass of solute} + \text{mass of solvent} = 58.44\ \text{g} + 500.\ \text{g} = 558.44\ \text{g}$$

Answer: a. 558.44 g

Content (fully worked out problems answers) continued...

(24) A student measures a sugar sample as 120 mg instead of 0.12 g. What is the correct mass in grams? Explain

a. 0.12 g

b. 1.20 g

c. 0.012 g

d. 12.0 g

Solution:

$$120 \text{ mg} \times \frac{1 \text{ g}}{1000 \text{ mg}} = 0.12 \text{ g}$$

Answer: a. 0.12 g, the answer must explain the conversion of 120 mg to 0.12 g, instead of assuming it is equal to 0.12 g.

Content (fully worked out problems & answers) continued...

(25) What is my name?

a. Brandon

b. Michael

c. James

d. Daniel

Answer: a. Brandon

Content (full worked out problems & answers) continued...

(26) What is the professor's name?

a. Dr. Smith

b. Dr. Johnson

c. Dr. T, Dr. Tramontozzi

d. Dr. Brown

Answer: c. Dr. T, Dr. Tramontozzi

Content (full worked out problems & answers) continued...

(27) What days is class and lab held?

a. Monday and Wednesday

b. Tuesday and Thursday

c. Friday only

d. Wednesday and Friday

Answer: b. Tuesday and Thursday

Content (full worked out problems & answers) continued...

(28) What time does class start and when does lab end?

a. 9:00 AM 12:00 PM

b. 8:00 AM 11:30 AM

c. 10:00 AM 1:30 PM

d. 1:00 PM 4:30 PM

Answer: b. 8:00 AM 11:30 AM

Content (full worked out problems & answers) continued...

(29) What days are the supplemental instruction sessions?

a. Mondays

b. Tuesdays

c. Fridays

d. Thursdays

Answer: b. Tuesdays

Content (full worked out problems & answers) continued...

(30) What times are the supplemental instruction sessions?

a. 9:00 AM 10:00 AM

b. 11:50 AM 12:50 PM

c. 2:00 PM 3:00 PM

d. 4:00 PM 5:00 PM

Answer: b. 11:50 AM 12:50 PM

Content (full worked out problems & answers) continued...

(31) What is OWL?

a. A campus tutoring service

b. Online Web Learning platform used for assignments and course resources

c. A student club

d. A type of exam

Answer: b. Online Web Learning platform used for assignments and course resources

Content (full worked out problems & answers) continued...

(32) What is Canvas?

a. A classroom whiteboard

b. A learning management system where grades and course materials are posted

c. A chemistry textbook

d. A scheduling app

Answer: b. A learning management system where grades and course materials are posted

Content (full worked out problems & answers) continued...

(33) What are the professor's office hours?

a. Monday 1011 AM only

b. Tuesday 11:30 AM 12 PM; Wednesday 7:308 AM, 11:302 PM, 56 PM; Thursday 11:30 AM 12 PM

c. Friday 13 PM only

d. By appointment only

Answer: b. Tuesday 11:30 AM 12 PM; Wednesday 7:308 AM, 11:302 PM, 56 PM; Thursday 11:30 AM 12 PM

Content (full worked out problems & answers) continued...

(34) Which room is the class in?

a. SB 210

b. SB 208

c. SB 105

d. SB 416

Answer: a. SB 210

Content (full worked out problems & answers) continued...

(35) Which room is the lab in?

a. SB 416

b. SB 210

c. SB 105

d. SB 310

Answer: a. SB 416

Content (full worked out problems & answers) continued...

(36) Which room is the supplemental instruction in?

a. SB 208

b. SB 210

c. SB 105

d. SB 416

Answer: a. SB 208

Content (full worked out problems & answers) continued...

(37) Which campus are we at?

a. North Campus

b. South Campus

c. East Campus

d. West Campus

Answer: b. South Campus

Content (full worked out problems & answers) continued...

(38) What does a ruler measure?

a. Mass

b. Length

c. Temperature

d. Volume

Answer: b. Length

Content (full worked out problems & answers) continued...

(39) What does a scale measure?

a. Mass

b. Temperature

c. Length

d. Pressure

Answer: a. Mass

Content (full worked out problems & answers) continued...

(40) What does a volumetric flask measure?

a. Approximate volume

b. Temperature

c. Precise volume of liquids

d. Mass

Answer: c. Precise volume of liquids

Content (full worked out problems & answers) continued...

(41) What does a thermometer measure?

a. Pressure

b. Temperature

c. Volume

d. Mass

Answer: b. Temperature

Content (full worked out problems & answers) continued...

(42) What does a graduated cylinder measure?

a. Mass

b. Volume of liquids

c. Temperature

d. Pressure

Answer: b. Volume of liquids

Content (full worked out problems & answers) continued...

(43) What does a pipette measure?

a. Approximate volume

b. Small, precise amounts of liquid

c. Mass

d. Temperature

Answer: b. Small, precise amounts of liquid

Content (full worked out problems & answers) continued...

(44) What does a beaker measure?

a. Exact volume of liquids

b. Approximate volume of liquids

c. Temperature

d. Mass

Answer: b. Approximate volume of liquids

Content (full worked out problems & answers) continued...

(45) What does a balance measure?

a. Volume

b. Mass of objects or substances

c. Temperature

d. Pressure

Answer: b. Mass of objects or substances

Content (full worked out problems & answers) continued...

(46) What are significant figures?

a. The digits in a number which follow strict rules about precision

b. All numbers written in scientific notation

c. Only the digits before the decimal point

d. Only the zeroes in a measurement

Answer: a. The digits in a number which follow strict rules about precision

Content (full worked out problems & answers) continued...

(47) What's the difference of significant figures betweenexact and inexact numbers?

a. Exact numbers have infinite significant figures; inexact depend on measurement device

b. Exact numbers have no significant figures; inexact always have three

c. Exact numbers are estimates; inexact are precise

d. Exact numbers are rounded; inexact numbers are not

Answer: a. Exact numbers have infinite significant figures; inexact depend on measurement device

Content (full worked out problems & answers) continued...

(48) What is the fastest achievable speed?

a. Speed of sound

b. Speed of light

c. Speed of electrons in an atom

d. Escape velocity of Earth

Answer: b. Speed of light, $c \approx 3.00 \times 10^8\,\mathrm{m/s}$

Content (full worked out problems & answers) continued...

(49) Which is more important in representing a number which is subtracted?

a. The answer must contain the same number of decimals of the lesser precise measurements.

b. The answer must contain the same number of decimals of the more precise measurements.

c. The answer must contain the same number of significant figures of the lesser precise measurement.

d. The answer must contain the same number of significant figures of the more precise measurements.

Answer: a. The answer must contain the same number of decimals of the lesser precise measurements.

Content (fully worked out problems & answers) continued...

(50) Convert blood glucose level of 125.0 mg/dL to g/L.

a. 0.0125 g/L

b. 0.125 g/L

c. 1.25 g/L

d. 12.5 g/L

Answer:

$$\text{Concentration} = \frac{125\ \cancel{\text{mg}}}{1\ \cancel{\text{dL}}} \times \frac{1\ \text{g}}{1000\ \cancel{\text{mg}}} \times \frac{10\ \cancel{\text{dL}}}{1\ \text{L}} = 0.125\ \text{g/L}$$

Content (fully worked out problems & answers) continued...

(51) Convert 0.085 m^3 to gallons. (Use 1 m^3 = 35.3147 ft^3, 1 ft^3 = 7.48052 gal)

a. 2.25 gal

b. 22.5 gal

c. 225 gal

d. 2250 gal

Answer:

$$\text{Volume} = \frac{0.0850\ \cancel{m^3}}{1} \times \frac{35.3147\ \text{ft}^3}{1\ \cancel{m^3}} \times \frac{7.48052\ \text{gal}}{1\ \cancel{ft^3}} = 22.5\ \text{gal}$$

Content (fully worked out problems & answers) continued...

(52) Lead has density 11.34 g/cm^3. What is the mass in kg of a 275.0 cm^3 sample?

a. 0.312 kg

b. 3.12 kg

c. 31.2 kg

d. 312 kg

Answer:

$$\frac{11.34 \text{ g}}{1 \cancel{\text{cm}^3}} \times \frac{275 \cancel{\text{cm}^3}}{1} = 3.12 \times 10^3 \text{ g}$$

$$\text{Mass} = \frac{3.12 \times 10^3 \cancel{\text{g}}}{1} \times \frac{1 \text{ kg}}{1000 \cancel{\text{g}}} = 3.12 \text{ kg}$$

Content (fully worked out problems & answers) continued...

(53) A tectonic plate moves 860 km per day. What is its speed in m/s?

a. 0.995 m/s

b. 9.95 m/s

c. 95.9 m/s

d. 0.0959 m/s

Answer:

$$\text{Speed} = \frac{860\ \cancel{\text{km}}}{\cancel{\text{day}}} \times \frac{1000\ \text{m}}{1\ \cancel{\text{km}}} \times \frac{1\ \cancel{\text{day}}}{86400\ \text{s}} = 9.95\ \text{m/s}$$

Content (fully worked out problems & answers) continued...

(54) Convert 750 mL to in^3. (Use 1 in = 2.54 cm)

a. 4.58 in^3

b. 45.8 in^3

c. 458 in^3

d. 0.458 in^3

Answer:

$$750 \text{ mL} = 750 \text{ cm}^3, \quad 1 \text{ in}^3 = (2.54 \text{ cm})^3 = 16.387 \text{ cm}^3$$

$$\text{Volume} = \frac{750 \text{ cm}^3}{16.387 \text{ cm}^3/\text{in}^3} = 45.8 \text{ in}^3$$

Content (fully worked out problems & answers) continued...

(55) Convert 88.0 kPa to atm. (Use 1 atm = 101.325 kPa)

a. 0.868 atm

b. 0.880 atm

c. 1.15 atm

d. 0.815 atm

Answer:

$$\text{Pressure} = \frac{88.0\ \cancel{\text{kPa}}}{1} \times \frac{1\ \text{atm}}{101.325\ \cancel{\text{kPa}}} = 0.868\ \text{atm}$$

Content (fully worked out problems & answers) continued...

(56) Convert 4.6 g/L to mg/dL.

a. 0.46 mg/dL

b. 4.6 mg/dL

c. 46 mg/dL

d. 460 mg/dL

Answer:

$$\frac{4.6\ \cancel{\text{g}}}{\cancel{\text{L}}} \times \frac{1000\ \text{mg}}{1\ \cancel{\text{g}}} \times \frac{1\ \text{dL}}{10\ \cancel{\text{L}}} = 46\ \text{mg/dL}$$

Content (fully worked out problems & answers) continued...

(57) A block has mass 245.0 g and dimensions 4.50 cm × 3.20 cm × 2.10 cm. Find its density in kg/m^3.

a. 8.10×10^2 kg/m^3

b. 8.10×10^3 kg/m^3

c. 8.10×10^4 kg/m^3

d. 8.10×10^5 kg/m^3

Answer:

$$V = (4.50)(3.20)(2.10)\ \text{cm}^3 = 30.2\ \text{cm}^3$$

$$\rho = \frac{245.0\ \text{g}}{30.2\ \text{cm}^3} = 8.10\ \text{g/mL} = 8.10\ \frac{\text{g}}{\text{cm}^3}$$

$$\text{Density} = \left(8.10\ \frac{\text{g}}{\text{cm}^3}\right) \times \frac{1\ \text{kg}}{1000\ \text{g}} \times \frac{10^6\ \text{cm}^3}{1\ \text{m}^3} = 8.10 \times 10^3\ \text{kg/m}^3$$

Content (fully worked out problems & answers) continued...

(58) Show that 125.0 N·m equals 125.0 J using unit analysis. (Use 1 N = 1 kg·m/s^2)

a. 125.0 J

b. 125.0 N

c. 125.0 W

d. 125.0 kg·m/s^2

Answer:

$$125.0\ \mathrm{N} \cdot \mathrm{m} = 125.0 \left(\frac{\mathrm{kg} \cdot \mathrm{m}}{\mathrm{s}^2} \right) \cdot \mathrm{m} = 125.0\ \frac{\mathrm{kg} \cdot \mathrm{m}^2}{\mathrm{s}^2} = 125.0\ \mathrm{J}$$

Content (fully worked out problems answers) continued...

(59) A student weighs a sample of water and records 248.5 g. The actual mass is 250.0 g. What is the difference between the recorded mass and the true mass?

Solution:

$$\text{Difference} = \text{True mass} - \text{Recorded mass} = 250.0\ \text{g} - 248.5\ \text{g} = 1.5\ \text{g}$$

Answer: 1.5 g

Content (fully worked out problems & answers) continued...

(60) A 0.500 g sample of NaCl dissolves in 250.0 mL of water. What is the concentration in g/L?

a. 0.125 g/L

b. 2.00 g/L

c. 1.25 g/L

d. 0.250 g/L

Answer:

$$\text{Concentration} = \frac{0.500 \text{ g}}{0.250 \text{ L}} = 2.00 \text{ g/L}$$

Content (fully worked out problems & answers) continued...

(61) Convert 987.0 mbar to Pa.

a. 9.870 Pa

b. 9.87×10^2 Pa

c. 9.87×10^4 Pa

d. 9.87×10^5 Pa

Answer:

$$987.0 \text{ mbar} \times \frac{100 \text{ Pa}}{1 \text{ mbar}} = 9.870 \times 10^4 \text{ Pa}$$

Content (fully worked out problems & answers) continued...

(62) Water has density 62.4 lb/ft^3. Convert to g/mL. (Use 1 lb = 453.592 g, 1 ft^3 = 28316.8467 mL)

a. 0.624 g/mL

b. 1.00 g/mL

c. 2.20 g/mL

d. 6.24 g/mL

Answer:

$$62.4 \text{ lb/ft}^3 \times \frac{453.592 \text{ g}}{1 \text{ lb}} \times \frac{1 \text{ ft}^3}{28316.8467 \text{ mL}} = 1.00 \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(63) A sample occupies 250.0 mL and has a mass of 0.890 g. Calculate its density.

a. 3.56×10^{-3} g/mL

b. 0.890 g/mL

c. 3.56 g/mL

d. 35.6 g/mL

Answer:

$$\rho = \frac{0.890 \text{ g}}{250.0 \text{ mL}} = 3.56 \times 10^{-3} \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(64) A beaker contains 3.50 L of solution. Express this volume in cubic decimeters (dm^3) using unit factors.

a. 0.350 dm^3

b. 3.50 dm^3

c. 35.0 dm^3

d. 350 dm^3

Answer:

$$3.50 \text{ L} \times \frac{1 \text{ dm}^3}{1 \text{ L}} = 3.50 \text{ dm}^3$$

Content (fully worked out problems & answers) continued...

(65) A student measures 0.00250 kg of a substance. Express this in grams.

a. 0.00250 g

b. 0.0250 g

c. 2.50 g

d. 250 g

Answer:

$$0.00250 \text{ kg} \times \frac{1000 \text{ g}}{1 \text{ kg}} = 2.50 \text{ g}$$

Content (fully worked out problems & answers) continued...

(66) A 5.00 mL sample of water weighs 4.98 g. What is its density?

a. 0.996 g/mL

b. 1.00 g/mL

c. 1.02 g/mL

d. 1.10 g/mL

Answer:

$$\rho = \frac{4.98 \text{ g}}{5.00 \text{ mL}} = 0.996 \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(67) Convert steel density 7850 kg/m^3 to lb/ft^3. (Use 1 m^3 = 35.3147 ft^3, 1 kg = 2.20462 lb)

a. 490.0 lb/ft^3

b. 785.0 lb/ft^3

c. 890 lb/ft^3

d. 354.0 lb/ft^3

Answer:

$$7850 \text{ kg/m}^3 \times \frac{2.20462 \text{ lb}}{1 \text{ kg}} \times \frac{1 \text{ m}^3}{35.3147 \text{ ft}^3} = 490.0 \text{ lb/ft}^3$$

Content (fully worked out problems & answers) continued...

(68) Multiply 3.445 cm $\times$ 2.1 cm.

a. 7.2 cm^2

b. 7.23 cm^2

c. 7.234 cm^2

d. 7.235 cm^2

Answer: 7.2 cm^2

$$3.445 \times 2.1 = 7.2345 \text{ cm}^2$$

With correct sig figs (2 sig figs, from 2.1): 7.2 cm^2

Content (fully worked out problems & answers) continued...

(69) Compute $\frac{4.500 \times 10^3 \text{g}}{2.6\text{L}}$. Report with correct sig figs.

a. 1.73×10^3 g/L

b. 1.7×10^3 g/L

c. 1.730×10^3 g/L

d. 1.75×10^3 g/L

Answer: 1.7×10^3 g/L

$$\frac{4.500 \times 10^3}{2.6} = 1730.7692 \text{ g/L}$$

With correct sig figs (2 sig figs, from 2.6 L): 1.7×10^3 g/L

Content (fully worked out problems & answers) continued...

(70) Add 2.05 g + 14.1 g + 3.222 g. Report with correct decimal-place precision.

a. 19.37 g

b. 19.372 g

c. 19.4 g

d. 19 g

Answer: 19.372 g

$$2.05 + 14.1 + 3.222 = 19.372 \text{ g}$$

Decimal-place precision: 19.372 g (all decimals retained)

Content (fully worked out problems & answers) continued...

(71) Subtract 4.500×10^2 mL from 1.200×10^3mL. Give the result with correct sig figs.

a. 7.50×10^2 mL

b. 7.500×10^2 mL

c. 7.5×10^2 mL

d. 750 mL

Answer: 7.500×10^2 mL

$$1.200 \times 10^3 - 4.500 \times 10^2 = 750 \text{ mL}$$

Correct sig figs (4 sig figs, matching 1.200×10^3 and 4.500×10^2): 7.500×10^2 mL

Content (fully worked out problems & answers) continued...

(72) Volume of a cube with side 3.44 cm.

a. 40.7 cm^3

b. 40.71 cm^3

c. 41 cm^3

d. 40 cm^3

Answer: 40.7 cm^3

$$V = (3.44)^3 = 40.707584 \text{ cm}^3$$

With correct sig figs (3 sig figs from 3.44): 40.7 cm^3

Content (fully worked out problems & answers) continued...

(73) Multiply 6.230×10^1cm $\times$ 4.0×10^{-2}cm. Give result with correct sig figs.

a. 2.492 cm^2

b. 2.49 cm^2

c. 2.5 cm^2

d. 2.50 cm^2

Answer: 2.5 cm^2

$$6.230 \times 10^1 \times 4.0 \times 10^{-2} = 2.492 \text{ cm}^2$$

Correct sig figs (2 sig figs, from 4.0): 2.5 cm^2

Content (fully worked out problems & answers) continued...

(74) A mass of 12.35 g and a mass of 1.2 g are combined. What is the combined mass with correct sig figs?

a. 13.55 g

b. 13.6 g

c. 13.5 g

d. 14 g

Answer: 13.55 g

$$12.35 + 1.2 = 13.55 \text{ g}$$

Decimal precision determined by 1.2 (2 sig figs): 13.55 g

Content (fully worked out problems & answers) continued...

(75) A length 4.80 km has an exact addition of 250 m. Express the total in km with correct sig figs.

a. 5.05 km

b. 5.050 km

c. 5.1 km

d. 5.0 km

Answer: 5.05 km

$$250 \text{ m} = 0.250 \text{ km}$$

$$4.80 + 0.250 = 5.05 \text{ km}$$

Content (fully worked out problems & answers) continued...

(76) Convert 9.60×10^1mph to m/s and report with correct sig figs.

a. 42.9 m/s

b. 43.0 m/s

c. 42.94 m/s

d. 42 m/s

Answer: 42.9 m/s

$$1 \text{ mph} = 0.44704 \text{ m/s}$$

$$96.0 \times 0.44704 = 42.892 \text{ m/s}$$

With 3 sig figs from 9.6010: 42.9 m/s

Content (fully worked out problems & answers) continued...

(77) A beaker has 125.0 mL; an exact 10.00 mL pipet adds more. What is the new volume with correct sig figs?

a. 135 mL

b. 135.0 mL

c. 135.00 mL

d. 135.000 mL

Answer: 135.0 mL

$$125.0 + 10.00 = 135.0 \text{ mL}$$

Correct sig figs: 135.0 mL (4 sig figs total)

Content (fully worked out problems & answers) continued...

(78) Convert 3.25 lb to grams using 1 lb = 453.6 g.

a. 1474 g

b. 1.47×10^3 g

c. 1.474×10^3 g

d. 1.48×10^3 g

Answer: 1.47×10^3 g

$$3.25 \times 453.6 = 1474.2 \text{ g}$$

Correct sig figs (3 sig figs from 3.25): 1.47×10^3 g

Content (fully worked out problems & answers) continued...

(79) Express 7.400×10^{-2}L in cm^3 (1 L = 1000 cm^3).

a. 74 cm^3

b. 74.0 cm^3

c. 74.00 cm^3

d. 74.000 cm^3

Answer: 74.00 cm^3

$$7.400 \times 10^{-2} \times 1000 = 74.0 \text{ cm}^3$$

Correct sig figs (4 sig figs from 7.40010): 74.00 cm^3

Content (fully worked out problems & answers) continued...

(80) You have 3.450×10^2 mL of solution and add 1.2×10^3 mL solvent. Total volume?

a. 1545 mL

b. 1.55×10^3 mL

c. 1.545×10^3 mL

d. 1.540×10^3 mL

Answer: 1.545×10^3 mL

$$345.0 + 1200 = 1545.0 \text{ mL}$$

Correct sig figs: 1.545×10^3 mL

Content (fully worked out problems & answers) continued...

(81) A bottle contains 1.003×10^3g of NaCl. Remove 125.0g. Mass remaining?

a. 878 g

b. 878.0 g

c. 877.95 g

d. 880 g

Answer: 878.0 g

$$1003 - 125.0 = 878.0 \text{ g}$$

Correct sig figs: 878.0 g

Content (fully worked out problems & answers) continued...

(82) Two glass rods: 12.45 cm and 8.0 cm. Combined length?

a. 20.45 cm

b. 20.5 cm

c. 20.4 cm

d. 21 cm

Answer: 20.45 cm

$$12.45 + 8.0 = 20.45 \text{ cm}$$

Correct sig figs (decimal place dictated by 8.0): 20.45 cm

Content (fully worked out problems & answers) continued...

(83) Stock solution volume 2.500×10^3mL. Add 45 mL water. Final volume?

a. 2545.0 mL

b. 2.545×10^3 mL

c. 2.55×10^3 mL

d. 2.54×10^3 mL

Answer: 2.545×10^3 mL

$$2500 + 45 = 2545 \text{ mL}$$

Correct sig figs: 2.545×10^3 mL

Content (fully worked out problems & answers) continued...

(84) Buret reading drops from 12.004 mL to 11.9965 mL. Subtract 0.0045 mL from 12.004. Report correct sig figs.

a. 11.9995 mL

b. 12.000 mL

c. 12.00 mL

d. 12.0 mL

Answer: 11.9995 mL

$$12.004 - 0.0045 = 11.9995 \text{ mL}$$

Correct sig figs (all digits meaningful): 11.9995 mL

Content (fully worked out problems & answers) continued...

(85) A metal foil width is 2.340 mm; coating adds 0.002 mm. Combined width?

a. 2.342 mm

b. 2.34 mm

c. 2.344 mm

d. 2.345 mm

Answer: 2.342 mm

$$2.340 + 0.002 = 2.342 \text{ mm}$$

Correct sig figs: 2.342 mm

Content (fully worked out problems & answers) continued...

(86) Wet sample mass 34.55 g; dry 33.1 g. Mass lost?

a. 1.45 g

b. 1.5 g

c. 1.50 g

d. 1.4 g

Answer: 1.45 g

$$34.55 - 33.1 = 1.45 \text{ g}$$

Correct sig figs: 1.45 g

Content (fully worked out problems & answers) continued...

(87) Reaction releases 4.50×10^4 J then 1.2×10^3 J. Total heat?

a. 4.62×10^4 J

b. 4.623×10^4 J

c. 4.6×10^4 J

d. 4.625×10^4 J

Answer: 4.62×10^4 J

$$4.50 \times 10^4 + 1.2 \times 10^3 = 46200 \text{ J} = 4.62 \times 10^4 \text{ J}$$

Content (fully worked out problems & answers) continued...

(88) Subtract 6.020×10^{23} molecules from 6.022×10^{23}. Difference?

a. 2.0×10^{20} molecules

b. 2×10^{20} molecules

c. 0.002×10^{23} molecules

d. 2.00×10^{20} molecules

Answer: 2.0×10^{20} molecules

$$6.022 \times 10^{23} - 6.020 \times 10^{23} = 0.002 \times 10^{23} = 2.0 \times 10^{20} \text{ molecules}$$

Content (fully worked out problems & answers) continued...

(89) Lab distances: 12.3 m and 8.45 m. Total?

a. 20.8 m

b. 20.75 m

c. 20.7 m

d. 21 m

Answer: 20.75 m

$$12.3 + 8.45 = 20.75 \text{ m}$$

Content (fully worked out problems & answers) continued...

(90) Add 0.00845 m and 0.045 m for a capillary. Sum?

a. 0.053 m

b. 0.05345 m

c. 0.0535 m

d. 0.054 m

Answer: 0.05345 m

$$0.00845 + 0.045 = 0.05345 \text{ m}$$

Content (fully worked out problems & answers) continued...

(91) You have 1.0000 L solvent; remove 1.00 L. Remaining volume?

a. 0.0000 L

b. 0.00 L

c. 0 L

d. 0.0 L

Answer: 0.00 L

$$1.0000 - 1.00 = 0.00 \text{ L}$$

Content (fully worked out problems & answers) continued...

(92) Combine metal rods: 0.452 m and 0.0035 m. Total length?

a. 0.456 m

b. 0.4555 m

c. 0.455 m

d. 0.457 m

Answer: 0.4587 m

$$0.452 + 0.0035 = 0.4555 \text{ m}$$

Correct sig figs (4 sig figs from 0.0035): 0.4555 m

Content (fully worked out problems answers) continued...

(93) Which state of matter has a definite shape and volume?

a. Solid

b. Liquid

c. Gas

d. Plasma

Answer: a. Solid

Content (fully worked out problems answers) continued...

(94) What is a chemical property?

a. A characteristic observed without changing the substance

b. A characteristic that describes how a substance reacts with other substances

c. The color of a substance

d. The mass of a substance

Answer: b. A characteristic that describes how a substance reacts with other substances

Content (fully worked out problems answers) continued...

(95) Which of the following is a physical change?

a. Rusting of iron

b. Burning wood

c. Melting ice

d. Baking a cake

Answer: c. Melting ice

Content (fully worked out problems answers) continued...

(96) Which particle is found in the nucleus of an atom?

a. Electron

b. Proton

c. Neutron

d. Both b and c

Answer: d. Both b and c

Content (fully worked out problems answers) continued...

(97) What is an element?

a. A substance made of two or more different atoms chemically bonded

b. A pure substance made of only one type of atom

c. A mixture of compounds

d. A solution

Answer: b. A pure substance made of only one type of atom

Content (fully worked out problems answers) continued...

(98) Which of the following is an example of a chemical change?

a. Boiling water

b. Dissolving sugar in water

c. Burning paper

d. Freezing juice

Answer: c. Burning paper

Content (fully worked out problems answers) continued...

(99) Which phase of matter has particles that move freely and fill the shape of their container?

a. Solid

b. Liquid

c. Gas

d. Plasma

Answer: c. Gas

Content (fully worked out problems answers) continued...

(100) Which of the following is a physical property?

a. Flammability

b. Reactivity with acid

c. Density

d. Oxidation

Answer: c. Density

Content (fully worked out problems answers) continued...

(101) Which state of matter takes both the shape and volume of its container?

a. Solid

b. Liquid

c. Gas

d. Plasma

Answer: b. Liquid

Content (fully worked out problems answers) continued...

(102) What is the smallest particle of an element that retains its chemical properties?

a. Molecule

b. Atom

c. Ion

d. Compound

Answer: b. Atom

Content (fully worked out problems answers) continued...

(103) Which of the following is a chemical property?

a. Boiling point

b. Color

c. Reactivity with water

d. Density

Answer: c. Reactivity with water

Content (fully worked out problems answers) continued...

(104) Which of the following is an example of a physical change?

a. Tarnishing of silver

b. Cutting paper

c. Burning coal

d. Rusting iron

Answer: b. Cutting paper

Content (fully worked out problems answers) continued...

(105) What is the charge of an electron?

a. Positive

b. Negative

c. Neutral

d. Variable

Answer: b. Negative

Content (fully worked out problems answers) continued...

(106) Which of the following is NOT a phase of matter?

a. Solid

b. Liquid

c. Gas

d. Solution

Answer: d. Solution

Content (fully worked out problems answers) continued...

(107) What is the center of an atom called?

a. Electron cloud

b. Nucleus

c. Proton

d. Neutron

Answer: b. Nucleus

Content (fully worked out problems answers) continued...

(108) Which of the following is an element symbol?

a. HO

b. CO

c. O

d. NaCl

Answer: c. O

Content (fully worked out problems answers) continued...

(109) Which of the following is an example of a chemical change?

a. Freezing water

b. Dissolving salt in water

c. Cooking an egg

d. Breaking a glass

Answer: c. Cooking an egg

Content (fully worked out problems answers) continued...

(110) Which particle determines the chemical identity of an element?

a. Electron

b. Neutron

c. Proton

d. All of the above

Answer: c. Proton

Content (fully worked out problems answers) continued...

(111) What is the main difference between a physical change and a chemical change?

a. Physical changes alter the composition; chemical changes do not

b. Physical changes do not alter the composition; chemical changes do

c. Both alter the composition

d. Neither alters the composition

Answer: b. Physical changes do not alter the composition; chemical changes do

Content (fully worked out problems answers) continued...

(112) Which of the following is an example of a gas at room temperature?

a. Oxygen

b. Iron

c. Water

d. Mercury

Answer: a. Oxygen

Content (fully worked out problems answers) continued...

(113) What is the basic unit of matter?

a. Atom

b. Molecule

c. Element

d. Compound

Answer: a. Atom

Content (fully worked out problems answers) continued...

(114) Which of the following is a mixture?

a. Salt (NaCl)

b. Sugar (CHO)

c. Air

d. Oxygen

Answer: c. Air

Content (fully worked out problems answers) continued...

(115) Which of the following describes an atom?

a. Smallest particle of a compound

b. Smallest particle of an element that retains its properties

c. A molecule made of different elements

d. A type of mixture

Answer: b. Smallest particle of an element that retains its properties

Content (fully worked out problems answers) continued...

(116) Which of the following is a property that can be observed without changing the chemical composition?

a. Flammability

b. Reactivity

c. Melting point

d. Oxidation

Answer: c. Melting point

Content (fully worked out problems answers) continued...

(117) Which phase of matter has a definite volume but no definite shape?

a. Solid

b. Liquid

c. Gas

d. Plasma

Answer: b. Liquid

Activity 1

Collaborative Learning Techniques	Assigned Discussion Leader
Learning Strategy	Family Feud
Time	12:10—12:30
Materials	Calculator, writing utensil, paper / tablet, conversion sheets
Lecture/Textbook/Old Plans/Slide References	Chapter 1: Basics of Matter, Chapter 2: Measurements

Definition

1. **Assigned Discussion Leader:** One person in the group is asked to present on a topic or review material for the group and then lead the discussion for the group. This person should not be the regular group leader.
2. **Family Feud:** This will work well as a test or quiz review for concepts and making connections between terms. Have students write down one term that comes to mind when a broad category is given. Create a list of the top 4 or 5 and use those as "answers" for the game. During the session, dividie studetns into two groups. Given the topic (broad category from the survey), students can decide wheher they want to play or pass. If they play, they alternate turns calling out a term they think is relevant. If they are correct, they get the points. If not, the next student attempts. Continue in this fashion until all terms are guessed and explained or three incorrect guesses are made. If three incorrect guesses occur, the other team tgets the chance to finish the topic.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. Students are organized into teams and reminded that the game covers significant figures, unit conversions, basics of matter, story problems, and fun chemistry facts.
2. Each team takes turns selecting a category and point value. The team has a set time (e.g., 3060 seconds) to discuss and provide an answer.
3. The SI leader monitors discussions, ensuring all students participate and encouraging teams to explain their reasoning before answering.
4. If a team answers incorrectly, other teams have the opportunity to discuss and steal the question.
5. Points are awarded for correct answers and tracked on the board or a visible scoreboard, helping students see progress and encouraging friendly competition.
6. Between rounds, the SI leader pauses to clarify any misconceptions, review key concepts, and provide real-life examples to connect the question to practical applications.
7. After all questions are answered, the SI leader leads a short debrief, asking teams to reflect on what concepts were challenging, what strategies helped, and how preparation and teamwork influenced performance.

Checking for Understanding (questions & answers)

Using a Jeopardy Game to Review Chemistry Concepts Engaging students in a Jeopardy-style game serves as an effective and interactive method for reviewing essential chemistry topics while simultaneously assessing comprehension. The benefits of this approach are multi-faceted:

- **Coverage of Key Topics:** The game includes questions on significant figures, unit conversions, basics of matter, story problems, and fun chemistry facts. This ensures that students revisit each major concept from the chapters, from understanding the proper number of digits in measurements to applying unit conversions in practical scenarios, such as calculating the volume of a liquid medication or determining the mass of a sample in a lab experiment.
- **Formative Assessment:** By participating in the game, students gain immediate feedback on their understanding. For example, if a team struggles to answer a question about converting grams to moles, they recognize that additional practice is needed before moving on to more advanced topics. This self-assessment mirrors real-life learning, where recognizing gaps in knowledge is the first step toward improvement.
- **Collaborative Learning:** Students work in teams, encouraging discussion, debate, and collective problem-solving. For instance, when faced with a story problem about mixing solutions, team members must communicate and decide which approach to take. This mirrors real-life scientific work, where collaboration and peer feedback often lead to better outcomes than working in isolation.
- **Preparation and Study Habits:** The game highlights the importance of preparation. Students who review class notes, practice calculations, and understand core concepts perform better. This reflects real-life situations, such as preparing for lab experiments or exams, where consistent preparation leads to confidence and success.
- **Engagement and Motivation:** Incorporating game elements makes learning enjoyable, helping students stay motivated. For example, earning points for correctly answering a question about chemical reactions can make abstract concepts more tangible, similar to using simulations or interactive apps to visualize chemical processes in the real world.
- **Reflection on Learning:** After the game, students can reflect on their performance. They can ask themselves questions like: Which topics were easy, which were challenging, and what strategies helped my team succeed? This reflection encourages metacognition, the real-world skill of thinking about ones own learning process, which is crucial in both academic and professional settings.

Activity 2

Collaborative Learning Techniques	Fishbowl (many), Jigsaw (few)
Learning Strategy	Boardwork Model (modified)
Time	12:30—12:50
Materials	Colored whiteboard markers, a whiteboard, partitioned questions (to display multiple to multiple groups at a time).
Lecture/Textbook/Old Plans/Slide References	Chapter 1: Basics of Matter, Chapter 2: Measurements

Definitions

1. **Fishbowl:** The group is separated into an "inside" and an "outside" the fishbowl. Students inside the fishbowl will participate in the activity or discussion. Students outside the fishbowl will listen to the discussion and observe the students engage in the activity. After the fishbowl, students inside are given the opportunity to reflect on what they observed, provide feedback to those isnide, and can contribute more to the discussion.

2. **Jigsaw:** Jigsaws, when used properly, make the group as a whole dependent upon all of them in subgroups. Each group provides a peice of the puzzle. Group members are broken into smaller groups. Each small group works on some aspect of the same problem, question, or issue. They then share their part of the puzzle with the large group.

3. **Boardwork Model:** This is a method of organizing boardwork in order to fascilitate an undertsanding of problem-solving strategies. The baord should be divdied into 4 sections: 1.) Prerequisite Knowledge, 2.) Mathematical Steps, 3.) Narrative of the Steps, 4.) Additional Sample Problem. Encourage one student to fill out section 1 on the board. Then encourage 2 students to simultaneously complete section 2 and 3 on the board. Lastly, have another student complete section 4. Encourage the students to use this model when studying outside of the SI session.

Instructions (please provide aDETAILED activity breakdown below (and/or attached)

1. **Objective:** Guide students to work in groups of at least 2 to solve chemistry-based problems collaboratively on a whiteboard.

Setup

- Arrange the classroom so multiple groups can work at the whiteboard simultaneously.
- Provide each group with markers and a section of the whiteboard.
- Prepare a set of chemistry problems in advance.
- Optionally, provide reference tables or hints for problem-solving to keep groups on track.

Running the Activity

1. Introduce the activity and explain that groups of at least 2 will calculate problem results together.
2. Present the first chemistry problem to all groups.
3. Groups collaborate to solve the problem on the whiteboard.
4. Rotate through problems, allowing each group to contribute their calculations and drawings.
5. Monitor group progress, providing hints or clarifications as needed without giving away solutions.
6. Encourage discussion and peer teaching within groups to reinforce chemistry concepts.
7. Continue until all problems are solved and the final picture is completed.

Facilitator Role

- Ensure groups are on task and collaborating effectively.
- Clarify instructions and answer conceptual questions without solving problems for the students.
- Track timing to ensure the activity progresses smoothly.
- Circulate to observe problem-solving strategies, noting areas where students struggle for future review.
- Celebrate the completion of the final picture and debrief with the class on the chemistry concepts applied.

Checking for Understanding (questions answers)

Questions:

1. Why does the boardwork model provide a calmer, low-pressure environment for students?
2. How does the centralized canvas in the boardwork model help reduce student anxiety about mistakes?
3. In what ways can students approach chemistry problems when using the boardwork model?
4. What types of chemistry tasks can students work through step by step with this model?
5. How do peers act as resources in the boardwork model?
6. How does peer explanation and feedback reinforce understanding of chemistry concepts?
7. Within a Supplemental Instruction (SI) context, how does the boardwork model encourage participation?
8. What benefits do students gain from multiple perspectives on the same chemistry problem?
9. How does the shared board help students practice applying theoretical knowledge in a visual way?
10. What role does the SI leader play when using the boardwork model during a session?
11. How does this model help the SI leader identify common misconceptions?
12. In what way does the boardwork model make SI sessions more effective for learners of varied skill levels?

Closer

Collaborative Learning Techniques	Group Survey
Learning Strategy	Brain dump
Time	12:50—12:55
Materials	Index cards, writing utensils
Lecture/Textbook/Old Plans/Slide References	Chapter 1: Basics of Matter, Chapter 2: Measurements

Definitions

1. **Group survey:** Each group member is surveyed to discover their position on an issue, problem or topic. This process ensures that each member of the group is allowed to offer or state their point of view.
2. **Brain dump:** This is a great closer for students to sum up their knowledge on the topic(s) from the session. Have the students take out a blank piece of paper /Google Doc and write/type("dump") everything they know relevant from the discussion that day. The SI Leader can look at their work to assess the students' understanding of the course material and have them compare it to each other. It is important to at least discuss with the students so the SI Leader knows the students' comfort levels with the material.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. Students are put into small groups and are asked to write down their confidence on a single notecard without names.
2. The cards are evidence of the group's confidence in the content.

Content (fully worked out problems & answers)

- No new worked problems; brain dump.

Checking for Understanding (questions & answers) continued...

- Review confidence levels
- Interpret the variety of confidence and their distribution in groups.

Plan Checklist

- ☐ Variety of Collaborative Learning Techniques and Learning Strategies
- ☐ Chapter/slide# where information is coming from
- ☐ Included timestamps (and buffer time as needed)
- ☐ Details on how leader will break up students
- ☐ Checking for Understanding questions
- ☐ Fully worked out problems and examples for each activity
- ☐ Necessary links required for online implementation (if applicable)
- ☐ Source material correctly cited (if applicable)
- ☐ Plan is uploaded to Teams